
Exercise 8

1 Explicit MPC

Compute the explicit MPC controller for the following

$$\begin{aligned} u^*(x) = \operatorname{argmin} \quad & 2u^\top u + (x + u)^\top H(x + u) \\ \text{s.t.} \quad & u \in [-1, 1] \end{aligned}$$

where H is the solution to the discrete-time algebraic Riccati equation for $A = 1$, $B = 1$, $Q = 1$ and $R = 2$.